Digital Design Verification

Exercise # 03

RISC-V – Procedures

Implementing Recursive Procedures

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NUST Chip Design Centre (NCDC), Islamabad, Pakistan

Revision History

Revision Number	Revision Date	Revision By	Nature of Revision	Approved By
1.0	30/05/2024	Muhammad Imran	Complete manual	-

Contents

Objective

The objective of this lab is to:

- Understand and analyze the implementation of a recursive procedure in RISC-V assembly language.

Tools

- Venus

Task 1

Implement the following recursive C language procedure in RISC-V assembly language.

```
int fact (int n)
{
        if (n < 1) return (1);
        else return (n * fact(n − 1));
}
```

- x10 = n (input argument)
- Returned value is also in x10
- May use x5 as temporary register!
- Multiply instruction:
  - **mul x10, x10, x6**
  - **x10 = x10 x x6**

Task 2

Implement the following recursive C language procedure in RISC-V assembly language.

```
int fib(int n){
    if (n==0)
        return 0;
    else if (n == 1)
        return 1;
    else
        return fib(n−1) + fib(n−2);
}
```

- x10 = n (input argument)
- Returned value is also in x10